SRINIVASAN T

Embedded Firmware Engineer
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Professional Summary

Embedded Firmware Engineer with hands-on experience in Embedded C/C++, firmware development, microcontroller programming, IoT systems, and embedded software testing. Proficient in ESP32, STM32, Arduino, UART, SPI, I2C, CAN, GPIO, Timers, Interrupts, RTOS concepts, Git, and Python. Strong analytical, debugging, and problem-solving skills with experience in hardware-software integration and firmware validation.

Technical Skills

Programming Languages: Embedded C, C++, Python
Microcontrollers: ESP32, STM32, Arduino UNO, Raspberry Pi
Communication Protocols: UART, SPI, I2C, CAN, MQTT, BLE, Wi-Fi
Embedded Concepts: GPIO, Timers, Interrupts, PWM, ADC, Firmware Development, Firmware Testing, Debugging, Hardware-Software Integration, RTOS (Basics)
Development Tools: STM32CubeIDE, PlatformIO, Arduino IDE, VS Code, Git, GitHub, MATLAB, Simulink

Professional Experience

Firmware Developer Intern Feb 2026 – May 2026
SSS Technologies

- Developed Embedded C firmware for automation and monitoring systems.
- Implemented GPIO, ADC, Timers, Interrupts and actuator control.
- Performed firmware debugging, testing and system validation.
- Worked on hardware-software integration and RTOS concepts.
- Prepared technical documentation and test reports.

Embedded Systems Intern Jul 2025 – Aug 2025
Zeekers Technology Solutions Pvt. Ltd.

- Developed embedded applications using ESP32.
- Implemented UART, SPI and I2C communication.
- Integrated sensors and actuators for IoT applications.
- Performed firmware debugging and testing.

Projects

Autonomous IoT Fire Extinguisher Robot

- Developed Embedded C firmware using ESP32.
- Integrated flame sensors, relay module and L298N motor driver.
- Implemented real-time fire detection and suppression logic.
- Used ThingSpeak for IoT monitoring and remote status updates.

Vision Guided Robotic Inspection System

- Developed AI-based defect detection system with 93% accuracy.
- Improved inspection efficiency through automated image processing.

IoT Women's Safety Device

- Developed embedded firmware for emergency alert device.
- Integrated GPS and GSM modules for real-time location tracking.

Education

Bachelor of Engineering in Electronics and Communication Engineering
Thanthai Periyar Government Institute of Technology, Vellore
CGPA: 8.34/10 Graduated: 2026

Achievements

- Second Prize – TEXATHON 3.0
- Second Prize – REVIVA'24 Project Expo
- Second Prize – TECHHARVEST Hackathon

Certifications

- NPTEL – Industrial IoT and Industry 4.0
- NPTEL – Computer Vision